

Module 4 Programming III (11/21)

 Publish

Java Swing Student Grade Manager Lab Assignment

Objective:

The objective of this lab assignment is to create a Java Swing application for managing student grades. Students are expected to use the provided skeleton code to implement functionalities for inputting midterm and final scores, calculating the average, determining the letter grade, and saving the data to a MariaDB database.

Task Description:

1. Database Setup (15 points):

- Implement the `createTableIfNotExists` method to create a table named `student_grades` with the specified columns.
- Ensure that the table includes columns for `id` (auto-incremented), `midterm_score`, `final_score`, and `average_grade`.

2. GUI Implementation (25 points):

- Implement the Swing GUI to include text fields for entering midterm and final scores, a button to calculate the grade, and a label to display the average grade.

3. Calculation and Display (30 points):

- Implement the `calculateAndDisplayGrade` method to calculate the average grade based on the entered midterm and final scores.
- Display the calculated average grade on the GUI.

4. Database Interaction (20 points):

- Implement the `saveToDatabase` method to insert the student's midterm and final scores along with the calculated average grade into the `student_grades` table.
- Handle database connection and statement execution appropriately.

5. Code Structure and Readability (10 points):

- Ensure the code follows good practices, including proper indentation and meaningful variable/method names.
- Include comments where necessary to explain the functionality of methods and sections of code.

Grading Rubric:

• Database Setup (15 points):

- Successful creation of the `student_grades` table with the correct structure: 5 points.
- Primary key, auto-increment settings, and appropriate data types: 5 points.
- Proper error handling for database operations: 5 points.

• GUI Implementation (25 points):

- Presence of GUI components (text fields, button, label): 5 points.
- Proper layout (e.g., GridLayout) and organization of GUI elements: 5 points.
- Event handling for button click: 5 points.
- User-friendly design and clear component labels: 5 points.
- Proper error handling for user inputs: 5 points.

• Calculation and Display (30 points):

- Successful calculation of average grade: 10 points.
- Display of the calculated average grade on the GUI: 10 points.
- Proper handling of invalid input (non-numeric): 5 points.
- Clear and user-friendly messages: 5 points.

• Database Interaction (20 points):

- Successful insertion of data into the `student_grades` table: 10 points.
- Proper error handling for database operations: 5 points.
- Efficient handling of database connections and statements: 5 points.

• Code Structure and Readability (10 points):

- Consistent indentation and formatting: 5 points.
- Meaningful variable/method names: 5 points.

Submission Guidelines:

- Submit the completed Java Swing Student Grade Manager code as a single file.
- Include comments explaining the purpose of each method and major sections of code.
- Provide a brief summary of any challenges faced during implementation.

**** Skeleton Code ****

```
import javax.swing.*;
import java.awt.*;
import java.awt.event.ActionEvent;
import java.awt.event.ActionListener;
import java.sql.*;

public class StudentGradeManager {

    private static final String DB_URL = "jdbc:mariadb://localhost:3306/student_grade_db";
    private static final String DB_USER = "your_username";
    private static final String DB_PASSWORD = "your_password";

    public static void main(String[] args) {
        createTableIfNotExists();

        SwingUtilities.invokeLater(() -> {
            JFrame frame = new JFrame("Student Grade Manager");
            frame.setSize(400, 200);
            frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);

            JPanel panel = new JPanel();
            frame.add(panel);
```

```
        placeComponents(panel);

        frame.setVisible(true);
    });
}

private static void placeComponents(JPanel panel) {
    // TODO: Implement GUI components (text fields, button, label)

    JButton calculateButton = new JButton("Calculate Grade");
    JLabel resultLabel = new JLabel("Average Grade: ");

    // TODO: Add components to the panel and set layout

    calculateButton.addActionListener(new ActionListener() {
        @Override
        public void actionPerformed(ActionEvent e) {
            // TODO: Implement calculation and display logic
        }
    });
}

private static void createTableIfNotExists() {
    try (Connection connection = DriverManager.getConnection(DB_URL, DB_USER, DB_PASSWORD);
        Statement statement = connection.createStatement()) {
        // TODO: Implement table creation logic
    } catch (SQLException e) {
        e.printStackTrace();
    }
}

private static void calculateAndDisplayGrade(String midtermScore, String finalScore, JLabel resultLabel) {
    // TODO: Implement calculation and display logic
}
```

```

private static String calculateLetterGrade(double average) {
    // TODO: Implement logic to calculate letter grade
    return "";
}

private static void saveToDatabase(double midterm, double finalExam, String grade) {
    try (Connection connection = DriverManager.getConnection(DB_URL, DB_USER, DB_PASSWORD);
        PreparedStatement preparedStatement = connection.prepareStatement(
            "INSERT INTO student_grades (midterm_score, final_score, average_grade) VALUES (?, ?, ?)")) {
        preparedStatement.setDouble(1, midterm);
        preparedStatement.setDouble(2, finalExam);
        preparedStatement.setString(3, grade);
        preparedStatement.executeUpdate();
    } catch (SQLException e) {
        e.printStackTrace();
    }
}
}

```

Points 100

Submitting a file upload

Due	For	Available from	Until
Nov 18, 2025	Everyone	-	Nov 18, 2025 at 11:59pm